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Comparative Approach on Machine Learning and Deep Learning Techniques based Diabetic Retinopathy Detection

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***Abstract*—Diabetic Retinopathy (DR) is a disease that damages retinal blood vessels and leads to blindness. The main issue involved with DR detection is that manual diagnosis process is taken maximum computational and sensitive to analysis diabetic part has been noise or overlapping. This paper discusses the taxonomy for effective performance in manual detection of DR involving three various types: Deep Learning (DL), Machine Learning (ML) and Optimization based techniques. This survey aims to detect DR using image processing to reduces time and effort required ML based techniques such as Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) classifiers for disease detection. Deep Convolutional Neural Network (DCNN) and Convolutional Neural Networks (CNNs) algorithms have demonstrated their potential as a diagnostic tool for detecting DR in Deep Learning (DL) methods. To enhance high-level accuracy of DR detection, utilized optimization approaches like the Grey Wolf Optimization (GWO) algorithm. The performance of the model is evaluated based on parameters such as such as accuracy, precision, F1-score, specificity and sensitivity are used in this survey compared results with various DL approaches for DR detection.**

Keywords—convolutional neural network, deep learning, diabetic retinopathy, machine learning and grey wolf optimization.

I. Introduction

Diabetes is indeed a rapidly growing global health concern. According to recent statistics, approximately 380 million people worldwide have diabetes mellitus (DM) and it is predict that prevalence of disease, maximum to an estimated 590 million by 2024 [1]. Early-stage detection and treatment of DR are crucial since it is a progressive disease, its severity number of exudates and types of lesions in fundus images. Many people believe that diabetic retinopathy is irreversible and that it only be slowed during early stages. [2]. Every diabetic person needs regular and annual retinal screenings to ensure early detection and timely treatment of diabetic retinopathy, especially if it's threatening their sight. The conventional methods for screening retinopathy involve fundus examination by ophthalmologists or retinal colour imaging [3]. DR consists of two major types: Non-Proliferative Diabetic Retinopathy (NPDR) and Proliferative Diabetic Retinopathy (PDR). In its early stages, DR is referred to as NPDR, which is further categorized into mild, moderate and severe stages. [4].

Diabetic Retinopathy is used in three stages: (I) mild NPDR, it was earliest stage of DR [5]. (II) moderate NPDR, and (III) severe NPDR, with PDR denoting the high stage of DR. However, there is still a lack of large-scale diabetic retinopathy detection and treatment systems for individuals. Automated techniques have been developed to identify retinal diseases using stained fundus images, aiming to address limitations of conventional diagnostic methods [6]. A system for detecting referable diabetic retinopathy utilizes a DCNN and automates segmentation of DR lesions by generating heat maps from convolutional layer, demonstrating potential for discovering new images. An adopted DL model called Inception V3 was used for detecting referable diabetic retinopathy based on development dataset. With extensive training data and expert grading of fundus images, system achieved impressive performance, attaining a high area under dataset [7].

The structure of the paper is as follows: section 2 describes taxonomy of Diabetic Retinopathy. Section 3 presents a comparative analysis. Section 4 regrading problem statement. Section 5 discuss the challenges. Section 5 discusses the conclusion.

II. Taxonomy for Diabetic Retinopathy

In this section, the taxonomy for effective performance in DR involved three types: Optimization-based GWO, ML-based SVM, KNN and ANN, DL-based DCNN, CNN and MCCNN, Hybrid approaches based SVM& RF and ANN&ELM in DR. Figure 1. Represents the taxonomy for DR detection.

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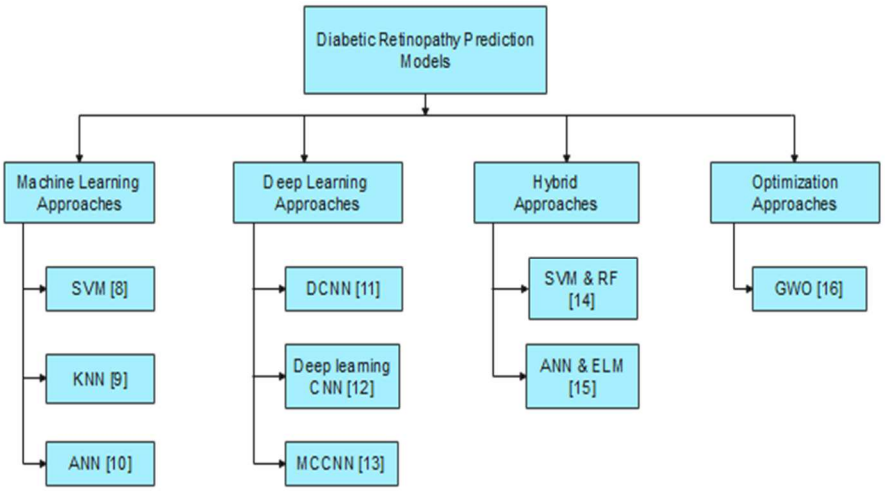

Fig. 1. Taxonomy for Diabetic Retinopathy detection

A. *Machine Learning Approaches*

Machine Learning (ML) approaches utilize various algorithms there are Support Vector Machine (SVM), K-Nearest Neighbour (KNN) classifier and Artificial Neural Network (ANN) to predict and achieve high accuracy in detection of diabetic retinopathy.

1) Support Vector Machine: Shubhi Gupta et al. [8] developed a ML approach that utilized SVM analysis for diagnosing diabetic retinopathy symptoms from fundus images. To analyse DR grading features, the ML technique for feature extraction. The SVM classifier's accuracy is significantly improved by utilizing a nonlinear kernel function, integrating advanced feature extraction and classification methods. The SVM classifier, employing the Haralick feature extraction method, achieves high performance and attains greater pixel accuracy. However, potential limitation is the need to establish high-quality datasets for validation, resulting in longer delays in obtaining detection results.

2) K-Nearest Neighbour Classifier: Ayushi Sharma et al. [9] presented a ML model for the detection of DR and KNN was used to analyse the dataset. This system aimed to classify images into tri categories: normal, diabetic retinopathy, and glaucoma using KNN. In this algorithm, a kernel function was utilized to assign weights to nearest k points. By involving weighted KNN algorithm high accuracy was achieved when combining exudates and blood vessel features. However, accuracy of KNN was slightly lower compared to existing system but still deemed satisfactory. The investigated used consistent data and combined features specifically for their analysis.

3) Artificial Neural Network Classifier: Ganeshsree Selvachandran et al. [10] developed a ML technique for detecting diabetic retinopathy, specifically using ANN. The supervised ML with ANN as a classification tool for DR. The technique used diverse numerical datasets to train ANN, resulting in high accuracy, resolution, while requiring a short time for diagnosis. Using these features as inputs in ANN proved beneficial for the early diagnosis of DR helping to prevent a prolonged and challenging treatment course and reducing the risk of rapid blindness.

B. *Deep Learning Approaches*

DL methods are used to enhance overall accuracy and reliability of DR detection with algorithms aimed at reducing generalization errors. Techniques such as Deep Convolutional Neural Network (DCNN), Convolutional Neural Network (CNN), and Multi-Class Convolutional Neural Network (MCCNN) are utilized for DR detection.

1) Deep Convolutional Neural Network: Richa Vij et al. [11] developed a DL based algorithm specifically DCNN model. This model training and validation uses a publicly accessible dataset with a small sample size. In experiments, various base pre-trained and DCNN-based models for multiclass DR severity stage classification using dataset. The classification achieved cutting-edge results, but further improvements were needed to accommodate larger datasets. The DCNN model was utilized to features extract, incorporating a global average pooling layer to mitigate overfitting. However, more complex to detect diabetics and a large number of parameters led to memory intensive model that took a long time to train dataset.

2) Convolutional Neural Network Classifier: Brijesh Kumar Chaurasia et al. [12] developed a CNN method for DR detection utilizing pre-trained models like DenseNet-169, VGG-19, and Inception-V3 on a large dataset alongside an ensemble learning strategy with models trained on the ImageNet dataset. However, running the dataset for an extensive number of epochs led to CNN model overfitting. To achieve optimal results, determine the right number of epochs and as a challenge, there are set it to 50 obtaining optimal results across all datasets. It also expanded our research to include federated learning and explored meta-heuristic optimization techniques. However, dealing with large datasets significantly increased computation time, so it chose to work with a limited dataset.

3) Multi-Channel Convolution Neural Network Classifier: M. Mohsin Butt et al. [13] developed a MCCNN for detecting DR in fundus images of eyes. To identify these lesions, several pixel classification and morphological-based

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extraction methods. These was handled multiclass classification for different stages of DR dataset and used deep MCCNN for pixel-wise exudate identified. The high resolution of input values in DR dataset resulted in a slowed training process, mainly due to memory issues caused by pixel size. There were conducted experiments with different epoch sizes as part of MCCNN approach. However, selection of appropriate design parameters involved impacted the model's performance. Detecting DR proved highly challenging due to variations in retina at different levels resulting in a lack of a linear relationship between epoch size and accuracy.

C. Hybrid Approaches

In Hybrid Approaches, efforts were made to design an automated system for detecting and diagnosing DR using hybrid learning features. Various techniques were used including DL-based SVM and ANN.

1) Support vector machine: Muhammad Mohsin Butt et al. [14] developed a DL-based SVM model to extract features and combine them to generate a hybrid feature. SVM classifies input data by creating a hyperplane in a high-dimensional space, allowing for application of a kernel function to transform input into hyperplanes, effectively separating data into distinct classes. SVM uses low level of classification process and aims maximum margins line between hyperplane classes. SVM classifier achieved high average class performance using hybrid feature vector for binary classification. However, sensitive to analysis diabetic part has been noise and dataset contains significant noise or overlapping target classes.

2) Artificial Neural Network: P. Preethy Rebecca and S. Allwin [15] developed an ANN to detect the DR in retinal image. This method included feature extraction followed by a classification stage using supervised classification methods. Automated DR systems were considered as alternatives to manual methods, significantly reducing the workload when applied to larger datasets. An automated system for extracting features in eye images is based on a tracking method that processes pixels. However, neural network scheme was used for pixel classification, and they used an extraction method but the vessel not maintain reliable performance.

D. Optimization Approaches

In optimization approaches, updated feature extractors using the nature-inspired Grey Wolf Optimization (GWO) algorithm based on a deep neural network classifier to detection of Diabetic Retinopathy.

1) Grey Wolf Optimization based Deep Neural Network Classifier: Manaswini Jena et al. [16] developed a deep neural network classifier based on GWO to obtain an optimized feature extractor. The GWO approach served as a high-level independent algorithm framework and a large-scale search for attaining optimal followed a hierarchical pattern. It was used to optimize Neural Network weights for feature refinement from network. However, error gradients due to classified diabetic such as gradient and parameter updating. By utilizing the GWO algorithm, these features were no longer reliant on classification errors.

III. Comparative Analysis

In this section, several DL and ML based techniques were established for DR detection and treatment. The comparative analysis includes author, methodology, advantages, limitation and performance metrics. Table 1 illustrates the explanation of existing DL and ML based method.

TABLE I. Comparative analysis of existing methods

Author	Method	Advantages	limitation	Performance matric
Shubhi Gupta et al. [8]	SVM	The SVM classifier with the Haralick feature extraction method, delivered high performance and achieved a higher level of pixel accuracy.	A potential limitation is that developing high-quality datasets for validation is required, and this resulted in higher delays in obtaining the detection results.	Accuracy, specificity, precision and sensitivity
Ayushi Sharma et al. [9]	KNN	DR-related to detect the vision is not accurately and lack of time.	The accuracy of KNN was slightly lower compared to the existing system but still deemed satisfactory. The investigated used consistent data and combined features specifically for their analysis.	Accuracy, specificity and sensitivity.
Ganeshsree Selvachandran et al. [10]	ANN	This technique achieves maximum resolution and specificity while requiring only a short time for diagnosis.	DR requires a large amount of labelled data for training in DR.	Accuracy, AUC and precision.
Richa Vij al et. [11]	DCNN	Screening numerous fundus images simultaneously demands a significant amount of time and is prone to diagnostic inaccuracies for DR.	They did not consider datasets to achieve greater generalizability, to expand to include acquired datasets. used balance class label was used in the dataset and utilized the pre-existing up-sampling technique.	AUC, Precision, Accuracy and Specificity.
Brijesh Kumar Chaurasia et al. [12]	CNN	The pixel with highest intensity value typically resides within optic cup, as this is a characteristic of retinal fundus image. This occurs when batch size is higher.	To address challenge, it set number of batches to 50, which produced optimal results with all datasets. To mitigate significantly prolonged computation times with the large dataset, used a limited dataset.	Accuracy, precision, Recall and F1-Score
M. Mohsin Butt et al. [13]	MCCNN	DR dataset have high accuracy form the input dataset. The performance improves with larger epoch sizes, and it's evident that the model achieves the highest accuracy.	The challenging task was variation retina to detection of DR depend upon disease stages.	Sensitive, precision, AUC and specificity

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Muhammad Mohsin Butt et al. [14]	SVM	The SVM classifies the data to forming a hyperplane. Because of the hyperplane dimension is occur high space.	Input data was large to detect the DR. the task will be high processing time. But small dataset is not enough information	Accuracy, precision, Recall and F-measure.
P. Preethy Rebecca and S. Allwin. [15]	ANN	The method significantly reduces the screening conducted over a larger dataset preformed effectively.	An automated system for extracting features in retinal images. A neural network scheme was used for pixel classification, and the vessel method does not introduce maintaining reliable performance.	Accuracy.
Manaswini Jena et al. [16]	GWO	The GWO algorithm was independent to utilized a large scale in optimization.	The current classification model relies on error gradients, which suffer from limitations such as dependence on gradients and parameter updates. This dependence was addressed by utilizing the GWO algorithm, ensuring that features are no longer contingent on classification. error.	Accuracy, Specificity, Precision, Recall, F1-Score and AUC

IV. Problem Statement

- Developing high-quality datasets for diabetic retinopathy detection validation because, leads to delays in obtaining results, hindering timely diagnosis.
- Balancing accuracy and computational efficiency poses a challenge, as KNN yield slightly lower accuracy compared to existing systems despite satisfactory performance.
- Achieving greater generalizability in diabetic retinopathy detection requires expanding datasets to include varied sources, which was overlooked in previous investigations.
- Efficiently handling large datasets for diabetic retinopathy detection is crucial to avoid significantly prolonged computation times without compromising accuracy.
- The dependence on error gradients in current classification models limits performance and efficiency, highlighting the need for alternative algorithms like GWO to mitigate this issue.

V. Discussion

This section, assessed four methods and ultimately confirmed the effectiveness of the model. There was applied all the models to various datasets and formed an ensemble classifier instead of training isolated models. In ML approaches, various algorithms such as SVM, KNN and ANN were used. They efficiently predicted DR grading using the method. DL method various techniques to detect DR, such as DCNN, CNN, and MCCNN. Hybrid approaches were designed to create an automated system for detecting and diagnosing DR. Various techniques, including Deep Learning SVM and ANN, were involved in this process. Optimization approaches were used to update feature extractors using the GWO-based deep neural network, resulting in efficient DR detection with high specificity.

VI. Conclusion

Diabetic Retinopathy harms the retinal blood vessels in diabetes patients. SVM has proven effective in classifying fundus retinal images into multiple classes. The potential limitation is challenge of developing a high-quality dataset for validation, leading to detection delays. While KNN had slightly lower accuracy compared to the existing system, it remained satisfactory. The CNN pixel with highest intensity value typically resides within optic cup, as characteristic of retinal fundus image. This DCNN-based model for multiclass DR severity stage classification using dataset. There method scope for improvement in quality of datasets, as they currently suffer from noisy images and inaccurate labels. Noise in labels comprises inconsistent information. Furthermore, DL techniques to locate affected areas and detect the multiclass various stages of DR in fundus images. The neural network scheme for pixel classification along with vessel extraction method operated without interference and demonstrated reliable performance.

References

[1] Khan, Z., Khan, F.G., Khan, A., Rehman, Z.U., Shah, S., Qummar, S., Ali, F. and Pack, S., 2021. Diabetic retinopathy detection using VGG-NIN a deep learning architecture. IEEE Access, 9, pp.61408-61416.

[2] Mohanty, C., Mahapatra, S., Acharya, B., Kokkoras, F., Gerogiannis, V.C., Karamitsos, I. and Kanavos, A., 2023. Using deep learning architectures for detection and classification of diabetic retinopathy. Sensors, 23(12), p.5726.

[3] Kalyani, G., Janakiramaiah, B., Karuna, A. and Prasad, L.N., 2023. Diabetic retinopathy detection and classification using capsule networks. Complex & Intelligent Systems, 9(3), pp.2651-2664.

[4] Gadekallu, T.R., Khare, N., Bhattacharya, S., Singh, S., Maddikunta, P.K.R. and Srivastava, G., 2023. Deep neural networks to predict diabetic retinopathy. Journal of Ambient Intelligence and Humanized Computing, pp.1-14.

[5] Majumder, S. and Kehtarnavaz, N., 2021. Multitasking deep learning model for detection of five stages of diabetic retinopathy. IEEE Access, 9, pp.123220-123230.

[6] Valizadeh, A., Jafarzadeh Ghoushchi, S., Ranjbarzadeh, R. and Pourasad, Y., 2021. Presentation of a segmentation method for a diabetic retinopathy patient's fundus region detection using a convolutional neural network. Computational Intelligence and Neuroscience, 2021, pp.1-14.

[7] Usman, T.M., Saheed, Y.K., Ignace, D. and Nsang, A., 2023. Diabetic retinopathy detection using principal component analysis multi-label feature extraction and classification. International Journal of Cognitive Computing in Engineering, 4, pp.78-88.

[8] Gupta, S., Thakur, S. and Gupta, A., 2023. Comparative study of different machine learning models for automatic diabetic retinopathy detection using fundus image. Multimedia Tools and Applications, pp.1-32.

[9] Sharma, A., Shinde, S., Shaikh, I.I., Vyas, M. and Rani, S., 2021, February. Machine learning approach for detection of diabetic retinopathy with improved pre-processing. In 2021 International conference on computing, communication, and intelligent systems (ICCCIS) (pp. 517-522). IEEE

[10] Selvachandran, G., Quek, S.G., Paramesran, R., Ding, W. and Son, L.H., 2023. Developments in the detection of diabetic retinopathy: a state-of-the-art review of computer-aided diagnosis and machine learning methods. Artificial intelligence review, 56(2), pp.915-964.

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[11] Vij, R. and Arora, S., 2023. A novel deep transfer learning based computerized diagnostic Systems for Multi-class imbalanced diabetic retinopathy severity classification. Multimedia Tools and Applications, pp.1-38.

[12] Ramesh, G.P., 2020, September. Probabilistic Analysis of Dynamic Power and Area in Network on Chip. In IOP Conference Series: Materials Science and Engineering (Vol. 925, No. 1, p. 012066). IOP Publishing.

[13] Butt, M.M., Latif, G., Iskandar, D.A., Alghazo, J. and Khan, A.H., 2019. Multi-channel convolutions neural network based diabetic retinopathy detection from fundus images. Procedia Computer Science, 163, pp.283-291.

[14] Butt, M.M., Iskandar, D.A., Abdelhamid, S.E., Latif, G. and Alghazo, R., 2022. Diabetic Retinopathy Detection from Fundus Images of the Eye Using Hybrid Deep Learning Features. Diagnostics, 12(7), p.1607.

[15] Preethy Rebecca, P. and Allwin, S., 2021. Detection of DR from retinal fundus images using prediction ANN classifier and RG based threshold segmentation for diabetes. Journal of Ambient Intelligence and Humanized Computing, 12(12), pp.10733-10740.

[16] Jena, M., Mishra, D., Mishra, S.P. and Mallick, P.K., 2023. A tailored complex medical decision analysis model for diabetic retinopathy classification based on optimized un-supervised feature learning approach. Arabian Journal for Science and Engineering, 48(2), pp.2087-2099.

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